

A Predictive Analysis of Fatal Traffic Crashes in Chicago

Overview

Transportation infrastructure profoundly influences the mobility, safety, and well-being of city residents. Chicago is a prime example of this concept, particularly centered on enhancing safety measures for pedestrians and automobiles alike. A city initiative labelled “Vision Zero” seeks to eliminate all motor vehicle accidents resulting in severe injury and death. Using the City of Chicago’s Data Portal I was able to analyze over 740,000 crash records from 2016-2022, resulting in the identification of fatality patterns. Through the development of predictive models I sought to inform strategies that could enhance road safety and reduce fatalities.

Data and Methodology



Our analysis utilized three primary datasets:

- **Crashes**: Details of each incident, including location, weather conditions, and crash type.
- **People**: Information on individuals involved, such as demographics and injury severity.
- **Vehicles**: Data on vehicles involved, including type and maneuver at the time of the crash.

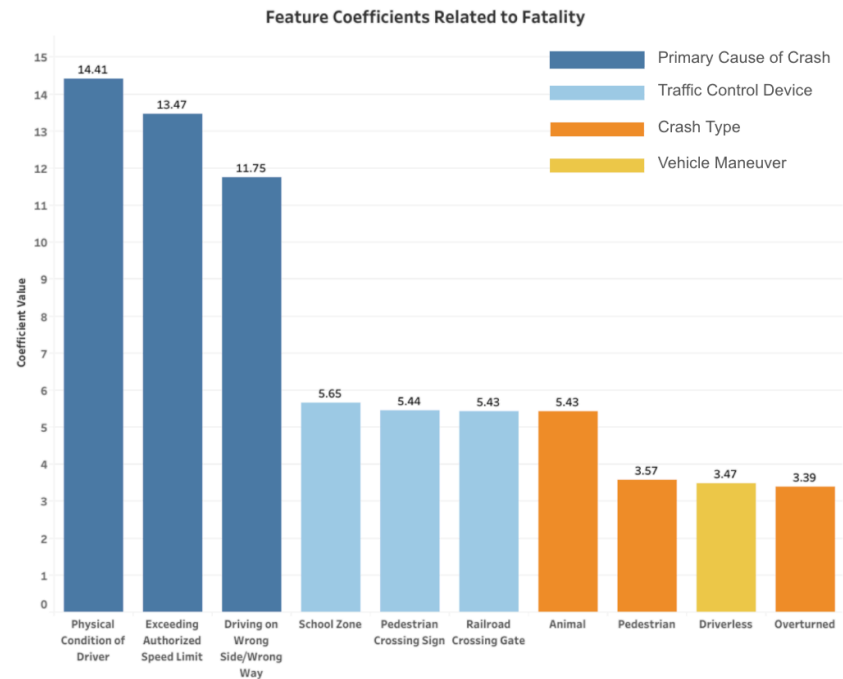
Most crashes in this dataset were not fatal—but about 1% of them did end tragically. To better understand these rare but serious incidents, the dataset was balanced using the *Synthetic Minority Over-sampling Technique (SMOTE)*, which creates artificial but realistic versions of fatal crashes based on patterns in the existing data. This helps the model learn to better recognize what leads to a fatal crash.

To include 14 categorical variables (e.g. weather condition, roadway surface condition, traffic device type, etc.) we used *one-hot encoding* - creating a binary column for each possible value. This expanded the dataset to 238 features the model could use during training.

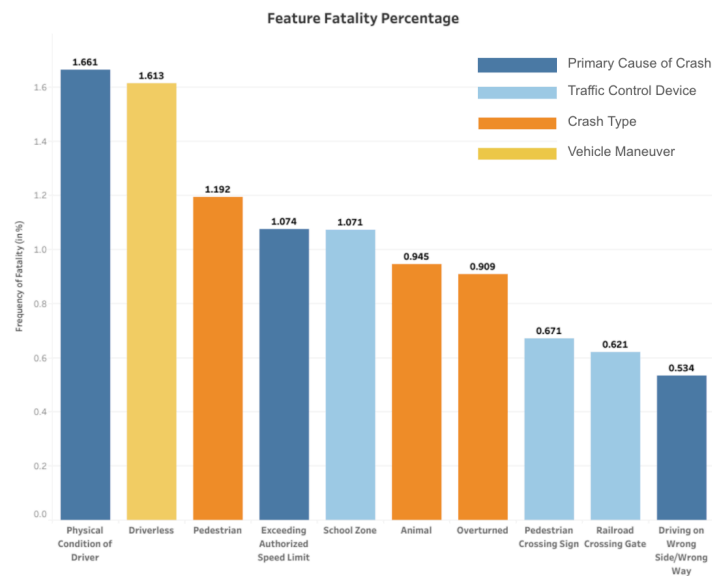
The main goal in developing a Logistic Regression and a Decision Tree model was to optimize the inclusion of fatalities resulting from car crashes. In prioritizing the identification of these fatalities, some false positives were likely included in this model, as there would be significant risk and danger in missing any true positives.

Model Performance and Key Findings

The Logistic Regression model outperformed the Decision Tree model and achieved a recall score of 0.70. Analysis of the model's coefficients highlighted several significant predictors of fatal crashes in four important categories: Primary Cause of Crash, Traffic Control Device, Crash Type, and Vehicle Maneuver.



The top predictors of a fatal crash were the physical condition of the driver (ie. impaired vs sober), followed by excessive speed, and driving on the wrong side of the road. By calculating the percentage of fatalities for each feature, I was able to decide which features to address resulting in the maximum impact



Addressing impaired driving requires broader societal and policy efforts, but there are infrastructure changes that could be implemented immediately to reduce fatal crashes caused by excessive speed, especially those pertaining to pedestrians.

Recommendations

Based on our findings, we propose two simple but powerful changes to how Chicago designs its streets to address two of the strongest predictors of fatal crashes: drivers exceeding the speed limit and crashes involving pedestrians.

- **Raised Crosswalks:** Elevating pedestrian crossings can lead to reduced vehicle speeds by 30 - 40%, reducing the chances of severe injuries.¹



- **Protected Bike Lanes:** Adding delineated bike lanes not only creates a physical buffer that protects cyclists from vehicle traffic, but also shortens the distance pedestrians must walk across moving lanes—reducing their exposure to danger. Research shows that streets with protected bike lanes see a 28% reduction in maximum vehicle speeds, making them safer for everyone.² Additionally, studies have found that narrowing travel lanes to make room for bike lanes typically does *not* lead to increased congestion, and can benefit all road users—drivers, cyclists, and pedestrians alike.³



These infrastructure changes are recommended by the U.S. Department of Transportation Federal Highway Administration and have been proven to reduce vehicle speeds and increase pedestrian safety.

Limitations

While our findings provide valuable insights, several limitations should be considered:

- **Infrastructure Constraints:** Although raised crosswalks and protected bike lanes have been shown to improve safety, implementing them citywide may be constrained by space, budget, and existing road layouts.
- **Model Tradeoffs and False Positives:** Because our primary modeling goal was to maximize recall, the model is more likely to predict a crash as fatal even when it is not (false positives). While this helps us identify more true fatal crashes, it also means some features may appear predictive due to overflagging non-fatal events as fatal.

Future Directions

In the future, to enhance road safety and address the remaining features that are highly predictable of a fatality, further analysis should include :

- **Geospatial Analysis:** Mapping crash data to identify high-risk areas for targeted interventions.
- **In-Depth Study of Driver Impairments:** Analyzing the specific causes of driver impairments to develop targeted prevention strategies.
- **Assessment of Autonomous Vehicles:** Evaluating the impact of self-driving technologies on traffic safety.

Conclusion

This data-driven approach provides valuable insights into the factors contributing to fatal traffic crashes in Chicago. By implementing targeted infrastructure improvements, focused on reducing vehicle speed and increasing pedestrian safety, such as raised crosswalks and protected bike lanes, the city could make significant strides toward achieving the Vision Zero goal of eliminating traffic fatalities and severe injuries.

Works Cited

1. "Campus walkability: enhancing pedestrian safety through installation of raised crosswalks." *IOP Conference Series: Earth and Environmental Science*, vol. 1347, 2024, p. 10. *iopscience*, <https://iopscience.iop.org/>.
2. "The Traffic Calming Effect of Delineated Bicycle Lanes." *Journal of Urban Mobility*, vol. 5, 2024, p. 10. *sciencedirect*, <https://www.sciencedirect.com/>.
3. *Proven Safety Countermeasures: Bicycle Lanes*. U.S. Department of Transportation, Federal Highway Administration, https://highways.dot.gov/sites/fhwa.dot.gov/files/Bicycle%20Lanes_508.pdf.

Note: Access the full analysis on GitHub [here](#).